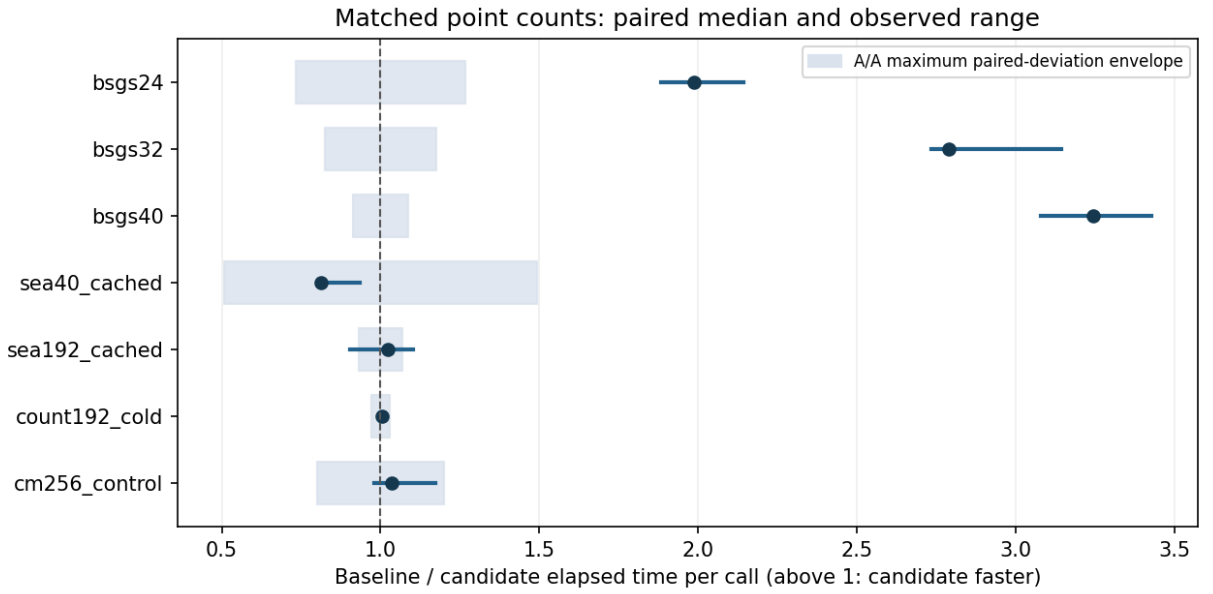
Matched point-count run 20261016T0205Z



run-20261016T0205Z; five A/B pairs per workload; pinned VM, PSI unavailable

Observed run, 2026-10-16 UTC. Original cf2745214d83e7bbf737f07d758e486d6af8fa7f versus candidate 44f8d17f1c5b8988a57a16e9dddc8e5ce099d325; native measurement source frozen at protocol commit bdcf3fca. Worker hashes and source identity are retained in frozen-hashes.txt, with Rust 1.99.1 (58cf8f9c3), release opt-level 3. The later CM generation fix is outside these timed revisions. It has separate untimed regression evidence.

Execution

Seven workloads, each five A/A pairs followed by five alternating A/B pairs: 140 requested records, 140 completed, exit 0, zero timeouts, zero mismatches, zero contention exclusions. A is always the original binary. An individual child has a 120-second limit; the suite has a 60-minute limit. Cache setup was prepared before the timed sequence. The original preparation recorded 82.296 seconds setup, separately from the measured cached-stage calls.

The host is an AMD EPYC 9V74 VM, Linux x86_64, five visible CPUs. The native runner holds /tmp/crypto-bench.lock, pins the child to CPU 4, evicts movable visible threads, observes process CPU use, and restores affinities. PSI is unavailable and retained as null. Unmovable system threads are recorded in host.json. Cloud neighbors and frequency are uncontrolled; no exclusive-host isolation is claimed.

Inputs and included work

Workload	Registered model	Calls per record
bsgs24	icv1-fp24-t6316-2e757e7a	1000
bsgs32	icv1-fp32-t6738-7a2b626e	300
bsgs40	icv1-fp40-tm1043-a68449f5	100
sea40_cached	icv1-fp40-tm1043-a68449f5	100
sea192_cached	icv1-fp192-t3160740231671392720748267719952e4af59	1
count192_cold	icv1-fp192-t3160740231671392720748267719952e4af59	1
cm256_control	icv1-fp256-t4324203865656596568524208663906731773276dadd18	10

Seeds are 1+i%4 within a record; one-call records use seed 1. Every result asserts the frozen registered order; complete API calls require both order certificates. The analyzer checks paired work, seeds, exact model and order, then independently certifies every reference after timing. This independent check shares the library arithmetic and probable-prime assumptions; it is not an independently verified arithmetic implementation.

BSGS rows time complete curve::order calls. Cached rows time sea::sea after canonical Phi-cache import and a warm-up; setup is separately recorded. Cold API counting includes fresh Phi construction, counting and certification. The unchanged CM count control uses the complete API on secp256k1. Validated input model construction, process launch and output serialization are excluded. Full accounting is in MEASUREMENT.md.

Native summary

Times are measured microseconds per call. Ratios and envelopes are derived by the native analyzer: median paired A/B ratio; observed paired minimum and maximum; maximum absolute A/A ratio deviation from one. There are five A/B pairs per row. An effect within its A/A envelope is inconclusive under the frozen policy. The envelope is descriptive, not a confidence interval.

Workload	A median us/call	B median us/call	A/B median ratio	Paired range	A/A max deviation	A/B pairs
bsgs24	61.747	31.495	1.98652	1.88259..2.14328	26.575%	5 / 5
bsgs32	327.251	117.994	2.79019	2.73356..3.14247	17.594%	5 / 5
bsgs40	1936.899	591.519	3.24310	3.07922..3.42764	8.710%	5 / 5
sea40_cached	264.650	314.525	0.81228	0.80232..0.93396	49.308%	5 / 5
sea192_cached	1904284.547	1825125.454	1.02393	0.90254..1.10098	6.766%	5 / 5
count192_cold	98449066.339	97670607.289	1.00469	0.99858..1.01812	2.869%	5 / 5
cm256_control	19048.878	17948.105	1.03405	0.98103..1.17123	19.957%	5 / 5

The native comparison admits the three BSGS effects beyond their A/A envelope. The other four effects are within it. Cached SEA40 has ratios below one in all five pairs. No complete-API or SEA improvement is established by this run.

Evidence and replay

runs.jsonl retains every observation, process accounting and output. summary.json retains the unrounded native statistics, including minimum arm times and setup medians. Reference certificates retain the independent post-timing order checks. No failed samples were omitted.

With Rust 1.99.1 installed, run bash replay-benchmark.sh from the study directory. It builds both frozen source worktrees, generates the excluded canonical cache, runs native reserve and analyze, and retains a fresh output directory. The original command used the baseline worker under /workspace/sea-original and the candidate worker under this study's native target/release, with output evidence/run-20261016T0205Z. Exact binary hashes are retained; rebuilding with the later constructor fix is not the timed arm.

The results report includes the formal checks and CM bug correction. Positive TLC/Lean results do not extend the measured workloads or prove the concrete Rust implementation. render_reports.py renders this report and its chart from the existing native summary without recomputing statistics or rerunning experiments.